

Supporting Information

MS²Agent: Multi-Agent LLM-Driven

Cross-Modal Alignment for Tandem Mass

Spectrometry Structural Annotation

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Appendix A: Framework Implementation, Training Data, and Model Details

A.1 Framework Data Flow and Intermodule Interactions

MS²Agent separates database-based tandem mass spectrometry (MS/MS) annotation into four modules: *MSpreprocessor*, *Structureextractor*, *Matcher*, and *Reranker*. The two report-generation modules use distinct input modalities. The *Matcher* evaluates their correspondence, and the *Reranker* revisits the selected candidates using the upstream reports.

The workflow uses serial module calls, tool-derived features, and retained intermediate reports.

These are generation constraints, not guarantees of factual correctness. A combined summary is formed from the MS and structure reports. Within each experimental configuration, the same LLM backbone is used for MS report generation, structure report generation, and reranker.

Formal data flow. Let X denote an experimental spectrum and $\mathcal{S} = \{S_i\}_{i=1}^N$ the candidate structure set. The report-generation stages are

$$R_{\text{MS}} = \text{MSpreprocessor}(X), \tag{S1}$$

$$R_{\text{str},i} = \text{Structureextractor}(S_i), \quad i = 1, \dots, N. \tag{S2}$$

The preliminary candidate list and selected subset are

$$C_{\text{pre}} = \text{Matcher}(R_{\text{MS}}, \{R_{\text{str},i}\}_{i=1}^N), \tag{S3}$$

$$C_n = \text{Top}_n(C_{\text{pre}}), \quad 1 \leq n \leq N. \tag{S4}$$

The final stage receives the selected candidate identifiers and their associated reports:

$$Y = \text{Reranker} (R_{\text{MS}}, \{(S_i, R_{\text{str},i}) : i \in C_n\}). \quad (\text{S5})$$

Here, $i \in C_n$ denotes membership in the selected candidate list. The specific logical data structure is shown in Figure S1.

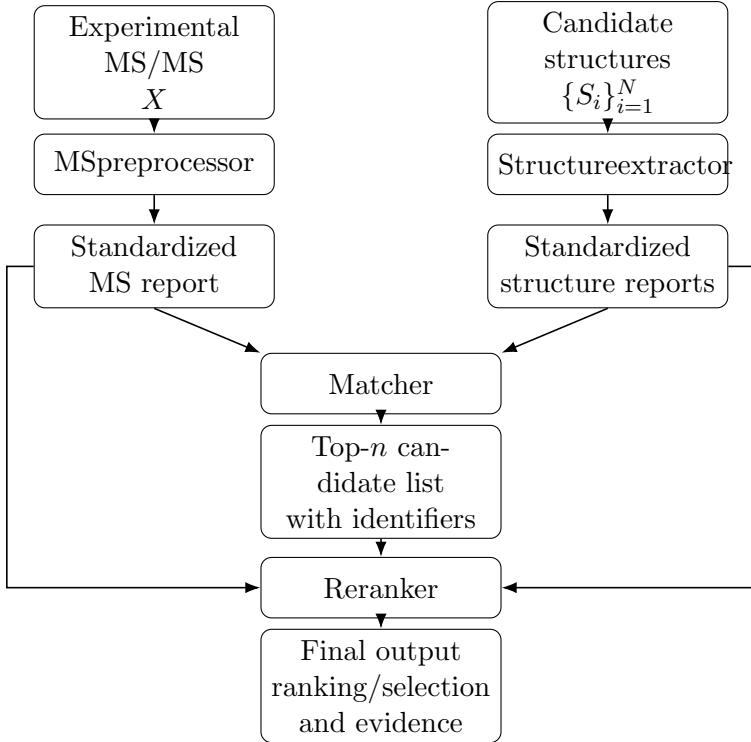


Figure S1: Documented information flow between the four modules. Arrows indicate input/output dependencies. The Reranker receives the spectral report and the selected candidates’ structural reports in addition to the preliminary candidate list.

Execution and state transfer. The serial processing path starts with `run_pipeline`, followed by `process_records` and `process_record_task`. Within a record, the input is normalized, spectral and structural features are prepared, and the MS and structure LLM calls are made in sequence. Their outputs are normalized and converted into summary text. After record processing, the workflow performs embedding, retrieval, and LLM reranking. The pseudocode below summarizes these stages.

Table S1: Module-level input and output dependencies.

Module	Input	Output
MSpreprocessor	Spectrum, available metadata, and computed peak summary	Standardized MS report
Structureextractor	Candidate SMILES, structural features, and structure image where used	Standardized structure report
Matcher	MS report and candidate structure reports	Preliminary scores
Reranker	Selected candidate identifiers, MS report, and associated structure reports	Choice and ranking outputs, confidence labels, and evidence tags

Input: records, candidate identifiers, reranking size n

- For each record:
 - Normalize the input record.
 - Run `summarize_peaks` and `infer_ms_mass_anchor`.
 - Obtain the structure image and structural features.
 - Build the MS and structure messages.
 - Obtain the cached or newly generated MS JSON response.
 - Obtain the cached or newly generated structure JSON response.
 - Normalize both responses and create summary text.
 - Retain the reports, identifiers, and available artifacts.
- Encode the summary texts and evaluate candidate matching.
- Select the top- n candidates from the preliminary ranking.
- Build reranking messages from query and candidate reports.
- Invoke both Choice and Ranking prompts for each query.
- Retain the resulting outputs for evaluation.

Shared-memory read/write. All intermediate artifacts are exchanged through a shared-memory layer rather than passed ad hoc. Within a record, the MS report R_{MS} and the per-candidate structure reports $R_{\text{str},i}$ are written by the MSpreprocessor and Structureextractor, respectively, and are read by the Matcher and Reranker; the preliminary scores and the Top- n list are written by the Matcher and read by the Reranker. To keep repeated calls deterministic, every LLM response is cached and keyed by the model identifier together with the complete prompt, PubChem properties are cached by CID, and structure images are

cached by SMILES hash. On a cache hit the stored JSON is replayed without recomputation; on a miss the response is generated and immediately written back to the cache.

Error correction. Before an output is forwarded downstream, the record applies the following deterministic exception handling, so that abnormal cases are corrected within a single record rather than by restarting the pipeline.

1. *Invalid report.* If a generated report fails JSON parsing or omits required fields, it is normalized, summarized, and the step is retried once.
2. *Insufficient evidence.* When spectral or structural evidence is missing, the affected field is emitted as **unknown** rather than fabricated; downstream embedding and reranking treat **unknown** as a missing feature.
3. *Unavailable structure image.* If the structure image cannot be retrieved, the system falls back from the PubChem PNG to RDKit rendering, then to a cached image, and finally marks it unavailable.
4. *Confidence normalization.* The free-text Reranker confidence is sanitized to the labels low/medium/high and is not interpreted as a calibrated probability.
5. *Cache miss.* An uncached LLM call is generated once and written back, so that subsequent reads reuse the stored response deterministically.

A.2 Training Data Construction

Data source. The training corpus was derived from a large-scale MS/MS–SMILES paired dataset constructed by integrating multiple public spectral resources. The original MS/MS spectra were collected from GNPS, MassBank, Spectraverse, and the SSFDB database from CSU-MS², supplemented with additional spectral datasets from our library. The corresponding molecular structures were obtained from public chemical databases and validated prior

to integration. After unified preprocessing, the dataset contained approximately 1.6 million experimental MS/MS–SMILES paired records, where each record links an experimentally acquired MS/MS spectrum to its corresponding molecular structure represented as a SMILES string.

The raw spectral data were first subjected to quality control, including precursor ion information completeness checking, fragment peak filtering, and removal of low-quality spectra. The associated molecular structures were subsequently standardized through unified structure representation, removal of salt-containing structures and duplicate structures, and molecular-level uniqueness verification based on InChIKey identifiers. During SMILES standardization, stereochemical information was retained to preserve structural specificity.

Training subset construction. For MS²Agent training, a high-quality molecular subset was selected from the above MS/MS–SMILES corpus according to predefined structural and spectral quality criteria. After filtering and quality assessment, 12,000 non-redundant molecular structures represented by standardized SMILES were retained as the MS²Agent training corpus. Each standardized molecule was associated with its corresponding experimental MS/MS spectra, resulting in the final MS/MS–SMILES training records.

This dataset was initially used to train a BERT-based similarity model; subsequently, 3,000 standardized molecules were sampled from the 12,000-molecule corpus for the optimization of MS2Rep-0.8B-GRPO. Sampling was performed at the SMILES molecule level to ensure that all MS/MS mass spectrometry data associated with a specific molecule were consistently assigned to the same training subset.

The resulting dataset supports spectrum-to-report and structure-to-report generation, as well as BERT-Sim training using matched and mismatched spectrum–structure report pairs. Details of BERT-Sim training are provided in Section A.3.2. BERT-Sim was trained using a supervised binary classification approach. Positive samples were constructed from matched "mass spectrum summary-structure summary" pairs found within the same exper-

imental record (label = 1), while negative samples were generated by randomly shuffling mass spectrum summaries and structure summaries across different records (label = 0). The ratio of positive to negative samples per batch was set to 1:3 to facilitate the calculation of losses for both sample types within the batch. Additionally, negative sampling incorporated candidate molecules from PubChem, utilizing random sampling with replacement for the negative samples in each batch.

Data leakage prevention and structural independence. To prevent potential data leakage, the training and evaluation datasets were separated according to predefined structural filtering criteria. For CASMI 2022 and the isomer benchmark dataset, compounds sharing the same 14-character InChIKey connectivity prefix with any training molecule were removed from the evaluation sets.

For MassSpecGym v1.5, a two-stage filtering strategy was applied. First, test molecules sharing the same 14-character InChIKey connectivity prefix with training molecules were removed. Subsequently, maximum common edge subgraph (MCES) distance-based filtering was performed to exclude any test molecule with an MCES distance of less than 10 to any training molecule, retaining only test molecules with an MCES distance of at least 10 from every training molecule.

After the two-stage filtering process, MassSpecGym v1.5 retained 4,295 query spectra covering more than 900 independent molecules for evaluation. To further quantify structural differences between training and evaluation datasets, pairwise Tanimoto similarities were calculated between training molecules and molecules from CASMI 2022 and the isomer benchmark dataset using Morgan fingerprints (radius = 2, 2048 bits). The maximum, mean, median, and distribution of Tanimoto similarity values are summarized in Table S2.

The average Tanimoto similarity across all three evaluation datasets is approximately 0.10, confirming that the training and test sets are structurally distinct at the population level. The majority of test molecules exhibit Tanimoto similarity below 0.15 to any training

Table S2: Pairwise Tanimoto similarity (ECFP4, 2048-bit) between the training set and each evaluation dataset.

Test set	Molecules	Mean Tanimoto	Median Tanimoto
MassSpecGym v1.5	967	0.0959	0.0909
CASMI 2022	497	0.1052	0.0935
Isomer benchmark	36	0.0971	0.0877

molecule, with fewer than 0.2% of all training–test pairs exceeding a Tanimoto threshold of 0.5. These results validate that model performance on these benchmarks reflects genuine generalization to unseen chemical structures rather than memorization of training examples.

A.3 Task-Specific Model Development

A.3.1 MS²Rep-0.8B-GRPO Training

MS²Rep-0.8B-GRPO was developed from Qwen3.5-0.8B using Group Relative Policy Optimization (GRPO). The model generates a standardized report conditioned on task input x and instruction c :

$$\hat{O} = G_{\theta}(x, c). \quad (\text{S6})$$

The two tasks use the same report-generation model with separate instructions for spectral and structural inputs. This is a task-adapted model, not a model evaluated without task-specific training.

Architecture and optimization. The base model is Qwen3.5-0.8B, trained using GRPO within the VERL framework. The policy objective employs a PPO clipping mechanism with a clipping ratio of 0.2 and incorporates a separate, low-variance KL divergence (KL loss) term relative to a frozen reference policy, with a coefficient of $\beta = 0.01$; this KL loss is excluded from the reward calculation.

The optimizer was AdamW with learning rate 1×10^{-6} , a constant learning-rate schedule, weight decay 0.01, gradient clipping at 1.0, and one PPO epoch per training step. Each step

used $B = 4$ prompts and $G = 6$ rollouts per prompt. Group-normalized reward advantages are defined as

$$\hat{A}_i = \frac{R(\hat{O}_i) - \mu_R}{\sigma_R + \epsilon}, \quad (\text{S7})$$

where μ_R and σ_R summarize rewards within the response group. A complete summary of hyperparameters is given in Table S4.

Reward function. The reward is specified as a programmatic combination of five binary rule-based criteria, accessed through `compute_score`:

$$R(\hat{O}) = \sum_{k=1}^5 w_k f_k(\hat{O}), \quad f_k \in \{0, 1\}, \quad \sum_{k=1}^5 w_k = 1. \quad (\text{S8})$$

Here, $\hat{O}$ denotes the generated report; each criterion applies the corresponding numerical or structural checks to that report. The reward specification is:

1. **Mass accuracy and ion-form inference** ($w_1 = 0.31$): rule-based scoring of mass deviation and mass/ion-form consistency.
2. **Fragment-ion coverage and spectrum profiling** ($w_2 = 0.29$): fragment-peak match counting with a matching tolerance of ± 0.5 Da. This is a training-reward tolerance, not a candidate-search mass window.
3. **Neutral-loss chemical plausibility** ($w_3 = 0.20$): matching reported neutral losses to common-loss patterns.
4. **Structural evidence grounding** ($w_4 = 0.14$): functional-group rule matching for the reported structural evidence.
5. **Output integrity** ($w_5 = 0.06$): checks of JSON field completeness.

The weights are fixed and empirically chosen: $w_1 = 0.31$, $w_2 = 0.29$, $w_3 = 0.20$, $w_4 = 0.14$, and $w_5 = 0.06$, summing to 1.

Training convergence. As shown in Figure S2(b), the smoothed group reward curve for MS²Rep-0.8B-GRPO exhibits a continuous upward trend throughout the displayed training phase. This gradual increase in reward values directly leads to higher-fidelity textual descriptions of MS/MS spectra and molecular SMILES, thereby providing richer discriminative features for the subsequent retrieval and re-ranking processes.

A.3.2 BERT-Sim Training

BERT-Sim was developed from BERT-base-uncased and trained on standardized MS-structure report pairs using binary labels $y_i \in \{0, 1\}$ for matched and mismatched pairs. The classifier output is

$$P_{\text{BERT},i} = \sigma \left(Wh_{[\text{CLS}]}^i + b \right), \quad (\text{S9})$$

where $h_{[\text{CLS}]}^i$ is the paired-input representation produced by the BERT-base-uncased encoder and σ is the sigmoid function. The model was optimized using binary cross-entropy loss for spectrum-structure consistency classification:

$$\mathcal{L}_{\text{BERT},i} = -y_i \log P_{\text{BERT},i} - (1 - y_i) \log (1 - P_{\text{BERT},i}). \quad (\text{S10})$$

Training configuration. The optimizer was AdamW with a learning rate of 5×10^{-5} and a batch size of 16. Training proceeded for 3 epochs. Hyperparameter details are listed in Table S4.

Validation. The development set was reused as the test input in the BERT training script; no separate held-out test partition was used within that script. The external evaluation datasets are described in Appendices C-E.

Training convergence. Figure S2(a) plots the training and test loss trajectories of BERT-Sim across 2000 optimization steps. Both loss values decline monotonically and plateau without late-stage divergence, maintaining a persistently low test loss throughout the training

schedule. The narrow, steady gap between training and test loss demonstrates that BERT-Sim captures generalizable spectrum–structure matching patterns instead of overfitting to training samples. This generalization capacity is essential for practical inference: BERT-Sim is required to score unseen spectrum–candidate pairs during library retrieval, and training-set memorization would severely degrade discriminative performance on out-of-distribution queries.

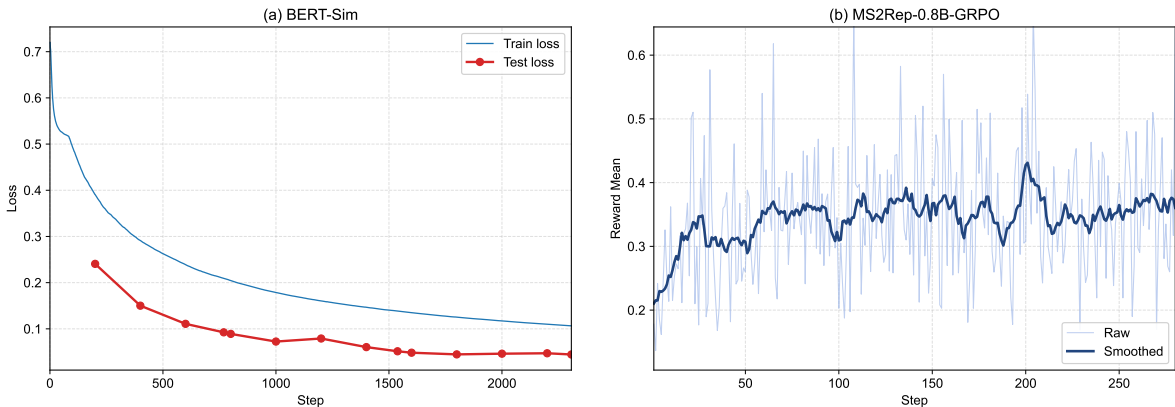


Figure S2: Optimization dynamics of core models. (a) Training and development loss curves of BERT-Sim. (b) Group-wise reward trajectory of MS²Rep-0.8B-GRPO.

A.4 Hardware and Software Environment

Within each experimental configuration, the MS report generator, structure report generator, and Reranker use the same LLM backbone. BERT-Sim remains a task-trained component in either setting.

Backbone identifiers and inference settings. The recorded backbone identifiers for the CASMI 2022 configurations include `gpt-5.4-mini-2026-03-17`, `gpt-5.4-nano-2026-03-17`, `qwen3.5-27b`, `qwen3.5-122b-a10b`, and `qwen3.5-0.8b`.

Hardware. GRPO training was performed on NVIDIA RTX 4080 SUPER GPUs with 32 GB memory.

Software. Table S3 lists the manually recorded software versions, CUDA/cuDNN entries describe the GPU environment.

Table S3: Manually recorded software environment.

Component	Version
Python	3.10
PyTorch	2.10.0
CUDA	12.8
cuDNN	8.9
Transformers	5.3.0
vLLM	0.18.0
RDKit	2023.09.5
NumPy	1.26.4
Pandas	3.0.3
scikit-learn	1.4.1

A.5 Data Interfaces and Availability

The interface accepts JSON-formatted MS/MS inputs with precursor m/z , charge, ionization mode, available metadata, and peak lists. It produces standardized MS and structure reports, candidate-ranking outputs, confidence labels, and analytical evidence. The public project repository is available at <https://github.com/JYresearch-group/MS2Agent>.

External evaluation datasets used in this study include CASMI 2022, MassSpecGym v1.5, and the isomer benchmark. Their construction and evaluation protocols are described in Appendices C, D, and E, respectively.

A.6 IMSS Specification and Report Templates

The Interpretable Mass Spectrometry Standard (IMSS) is the study-defined structured reporting specification used by MS²Agent. Here, a *standardized report* denotes a compact machine-readable retrieval signature, not an unrestricted narrative report.

Table S4: Training hyperparameters for the two task-specific models.

Component	Parameter	Value
MS ² Rep-0.8B-GRPO	Base model	Qwen3.5-0.8B
	Optimization	GRPO
	Optimizer	AdamW
	Learning rate	1×10^{-6}
	Batch size (prompts per step)	4
	Rollout group size	6
	KL coefficient	0.01
	Weight decay	0.01
	Gradient clipping	1.0
	PPO epochs	1
	Max new tokens	1,024
BERT-Sim	Base Model	BERT-base-uncased
	Optimizer	AdamW
	Learning rate	5×10^{-5}
	Batch size	16
	Epochs	3
	Loss	Binary cross-entropy
	Max sequence length	512
	Weight decay	0.0
	Warmup steps	50
	Adam epsilon	1×10^{-8}
	Max gradient norm	1.0
	Gradient accumulation steps	1
	Negative sampling ratio	1:3 (in-batch)

Numerical representation and vocabulary. Relative intensity is calculated as $100.0 I_j / I_{\text{base}}$ and expressed on a $[0, 100]$ scale, with the base peak at 100.0. The preprocessing representation retains m/z values to six decimal places; summary text uses four decimal places. Neutral-loss values are retained to four decimal places. A peak is represented by the fields `mz`, `intensity`, and `relative_intensity`.

The precursor-to-fragment difference is

$$\Delta(m/z)_j = (m/z)_{\text{precursor}} - (m/z)_j. \quad (\text{S11})$$

This calculation does not use charge or adduct information; the `neutral_losses` field therefore records precursor-to-fragment m/z differences without charge/adduct correction. Values are retained within $0.5 \leq \Delta(m/z)_j \leq 300$ Da. The precursor matching tolerance in the preprocessing code is 0.01 Da. A separate candidate mass-filter parameter of 0.05 Da is defined in the pipeline, but the retrieval path uses precomputed candidate-index files rather than the commented-out mass-mask operation. Numerical display precision, preprocessing tolerances, and candidate-search constraints are distinct.

MS report fields. Table S5 summarizes the nested MS report fields. Enumerated values are requested alternatives for a field, not a single pipe-delimited output string.

Table S5: Fields documented for the standardized MS report.

Parent	Field (type)	Meaning / values
Root	<code>analysis_type</code> (string)	<code>ms_retrieval_signature</code>
<code>mass_anchor</code>	<code>likely_ion_forms</code> (list)	Hypothesized adduct/ion forms
<code>mass_anchor</code>	<code>neutral_mass_hypotheses</code> (list)	Neutral-mass hypotheses associated with the ion-form interpretation
<code>spectrum_profile</code>	<code>precursor_dominance</code> (string)	<code>very_high</code> , <code>high</code> , <code>medium</code> , <code>low</code> , <code>unknown</code>
<code>spectrum_profile</code>	<code>fragmentation_richness</code> (string)	<code>very_low</code> , <code>low</code> , <code>medium</code> , <code>high</code> , <code>unknown</code>
<code>spectrum_profile</code>	<code>top_fragments_mz</code> (list)	Representative fragment m/z values
<code>spectrum_profile</code>	<code>neutral_losses</code> (list)	Precursor-to-fragment differences
Root	<code>chemical_hints</code> (list)	Compact chemical interpretation tags
Root	<code>retrieval_terms</code> (list)	3–6 retrieval-oriented terms

Structure report fields. Table S6 describes the structure-report template.

Table S6: Fields documented for the standardized structure report.

Parent	Field (type)	Meaning / values
Root	analysis_type (string)	structure_retrieval_signature
Root	consistency_check (string)	consistent, partially_consistent, uncertain, inconsistent
structure_profile	scaffold (string)	Scaffold description
structure_profile	ring_system (string)	Ring-system description
structure_profile	heteroatom_signature (string)	Encoded heteroatom counts
structure_profile	functional_groups (list)	Functional groups
structure_profile	acid_base_class (string)	strong_acid, weak_acid, neutral, basic, zwitterionic, unknown
structure_profile	shape_class (string)	fused_aromatic, planar_aromatic, flexible_aliphatic, mixed, unknown
fragmentation_priors	likely_losses (list)	Structure-derived loss labels
fragmentation_priors	fragile_bonds (list)	Structure-derived bond-cleavage hypotheses
Root	retrieval_terms (list)	Retrieval-oriented structural terms

Serialization templates. The skeletons below show the existing nested organization. Angle-bracket strings are placeholders, and empty lists mark fields to be populated; these skeletons are format illustrations, not measured examples or in-context demonstrations.

```
{
  "analysis_type": "ms_retrieval_signature",
  "mass_anchor": {
    "likely_ion_forms": ["<ion_form>"],
    "neutral_mass_hypotheses": []
  },
  "spectrum_profile": {
    "precursor_dominance": "<allowed_value>",
    "fragmentation_richness": "<allowed_value>",
    "top_fragments_mz": [],
    "neutral_losses": []
  }
}
```

```

    },
    "chemical_hints": ["<chemical_hint>"],
    "retrieval_terms": ["<retrieval_term>"]
  }

  {
    "analysis_type": "structure_retrieval_signature",
    "consistency_check": "<allowed_value>",
    "structure_profile": {
      "scaffold": "<scaffold_label>",
      "ring_system": "<ring_system_label>",
      "heteroatom_signature": "<heteroatom_counts>",
      "functional_groups": ["<functional_group>"],
      "acid_base_class": "<allowed_value>",
      "shape_class": "<allowed_value>"
    },
    "fragmentation_priors": {
      "likely_losses": ["<loss_label>"],
      "fragile_bonds": ["<bond_label>"]
    },
    "retrieval_terms": ["<retrieval_term>"]
  }

```

A.7 Report-Generation Prompts and Tool Chains

A.7.1 MSpreprocessor Prompts and Tool Chain

The MSpreprocessor uses the experimental spectrum and available instrumental metadata to generate the standardized MS report. Its prompt excludes candidate SMILES, structure images, database entries, and compound names from this generation stage.

System prompt.

System Prompt:

“You are a feature annotation expert for a mass spectrometry retrieval system. Your task is not to write an analytical report, but to convert instrumental results into stable, short, comparable retrieval signatures.

*Constraints: (1) Use only instrumental fields and **peak_summary** from the input. (2) Do not reference SMILES, structure images, database entries, or specific compound names. (3) If evidence is insufficient, output **unknown**; do not fabricate. (4) Do not output explanatory long text or Markdown. (5) Return JSON only. (6) Prioritize evidence that aids structure retrieval: mass anchors, representative fragments, neutral losses, spectrum sparsity.”*

User-prompt content. The user prompt requests conservative mass anchors, retrieval-relevant peak patterns, short chemical-hint tags, and 3–6 retrieval terms aligned with the structure-side vocabulary.

Spectral tool chain. `summarize_peaks` expresses relative intensities on [0,100] and selects representative peaks using a top- k procedure (default $k = 25$). The output fields are documented in Table S5.

`infer_ms_mass_anchor` prepares precursor-related mass anchors, likely ion-form hypotheses, and corresponding neutral-mass hypotheses from the available input information. Ionization-mode and instrument metadata are included where available. The numerical precursor-to-fragment difference calculation is separate from these ion-form hypotheses and does not apply charge/adduct corrections.

A.7.2 Structureextractor Prompts and Tool Chain

The Structureextractor generates a standardized structure report from candidate SMILES, structural features, and a two-dimensional structure image where used. This generation stage

is independent of the query MS report and does not use its experimental spectral fields.

System prompt.

System Prompt:

“You are a feature annotation expert for a chemical structure retrieval system. Your task is not to write a chemical review, but to convert SMILES and 2D structure images into stable, short, comparable retrieval signatures.

Constraints: (1) Use only SMILES, 2D structure images, and deterministically computed structural features from the input. (2) Do not reference experimental mass spectrometry fields or external database conclusions. (3) Perform a consistency check before outputting short retrieval tags. (4) Do not output explanatory long text or Markdown. (5) Return JSON only. (6) Prioritize structural evidence useful for mass spectral matching: mass anchors, functional groups, acid–base properties, fragile bonds, and typical neutral losses.”

User-prompt content. The user prompt requests preservation of tool-derived features, short labels for scaffold, ring system, functional groups, and fragile bonds, common neutral-loss labels, and 3–6 retrieval terms aligned with the MS-side vocabulary.

Structure preparation and tools. Structures are standardized and canonicalized before report generation. Training data and retrieval candidates use the same `build_structure_features` function. Candidate retrieval procedures for CASMI 2022, MassSpecGym, the isomer benchmark, and experimental validation are described in Appendices C, D, E, and F, respectively.

`get_structure_image` first attempts to obtain a PubChem PNG depiction, with fallback to RDKit rendering, a cached image, or an unavailable result. Images are keyed by a SMILES hash and are independent of the MS query.

`build_structure_features` retrieves PubChem properties including molecular formula, exact mass, hydrogen-bond donor and acceptor counts, and XLogP. It also supplies the heteroatom signature, XLogP class, and adduct compatibility based on SMILES-substring

acid/base patterns. PubChem property records are cached by CID. Functional-group and fragile-bond descriptions are generated by the LLM, rather than by a SMARTS-based recognition routine or a bond-dissociation-energy calculation in this tool.

The resulting structural inputs are supplied to the LLM. Its response is processed by `normalize_structure_analysis` and `make_structure_summary_text`. The structure-report field inventory is given in Table S6. Structure features and images do not depend on the query spectrum; the LLM-response cache is keyed by the model identifier and complete messages, as described in Section A.1.

A.8 Matcher Implementation and Score Combination

The Matcher combines Qwen3-Embedding report embeddings with the BERT-Sim binary classifier. It is a numerical inference stage rather than a separately prompted generative LLM. For the MS report embedding V_{MS} and a structure report embedding $V_{\text{str},i}$, cosine similarity is computed by L2 normalization followed by a vector dot product:

$$\text{Sim}(V_{\text{MS}}, V_{\text{str},i}) = \frac{V_{\text{MS}} \cdot V_{\text{str},i}}{\|V_{\text{MS}}\|_2 \|V_{\text{str},i}\|_2}. \quad (\text{S12})$$

The BERT-Sim classifier output $P_{\text{BERT},i}$ is defined in Equation S9. The preliminary matching score is

$$s_i = \alpha \text{Sim}(V_{\text{MS}}, V_{\text{str},i}) + (1 - \alpha) P_{\text{BERT},i}. \quad (\text{S13})$$

The weighting coefficient α is configurable, with a default value of 0.5. The numerical score is used to form the preliminary candidate ordering. Benchmark-specific candidate sets and reranking cutoffs are specified in Appendices C–E.

A.9 Reranker Decision Protocol and Output Handling

Two-stage score integration and conflict resolution. The Matcher score (Eq. S13) fuses the embedding-cosine and BERT-Sim signals into a single convex value s_i , but this

value is used only as an admission gate: it selects the Top- n candidates that are passed to the Reranker and does not determine their final order. The Reranker introduces no third numerical term and does not recompute s_i ; it re-decides both the order and the choice of the retained candidates using the evidence-priority ladder below. When a candidate is ranked high by s_i but fails one of the early evidence criteria—in particular criterion 1, exact-mass and adduct consistency—the Reranker demotes or rejects it, so that numerical similarity never overrides hard spectral consistency. Multi-dimensional matching is therefore resolved as a two-stage decision (numerical screening followed by evidence-based override) rather than as a single merged scalar.

Evidence-priority criteria. Reranking follows the priority order below:

1. Exact-mass and adduct consistency.
2. Neutral-loss and representative-fragment consistency.
3. Acid–base properties, heteroatom composition, and functional-group compatibility with the ionization mode.
4. Fragile-bond and likely-loss consistency.
5. Broad scaffold or chemical-family similarity.

Ranking mode. Ranking mode requests an ordering of the supplied candidate list. The prompt requires a complete permutation of the input identifiers without duplication. Its output fields are:

```
{
  "ranked_record_ids": ["<candidate_id_1>", "<candidate_id_2>"],
  "top_choice_record_id": "<candidate_id_1>",
  "confidence": "<low_or_medium_or_high>",
  "ranking_tags": ["<evidence_tag>"]
```

}

The prompt requests `low`, `medium`, or `high` confidence. The returned text is normalized using `sanitize_short_text`; it is not enum-validated or calibrated as a probability.

Appendix B: Scoring Criteria for Report Generation

Table S7: Scoring criteria for MS/MS report generation quality assessment (full score: 100).

Dimension Category	Scoring Item	Weight (%)	Core Definition	Scoring Criteria
Basic Peak	Peak filtering	10	Distinguish informative spectral peaks from low-confidence or background/noise signals	5: Correctly prioritizes major diagnostic or informative peaks based on spectral evidence while distinguishing important signals from low-confidence or noise peaks; 3: Identifies some major or high-intensity peaks but lacks systematic prioritization; 1: Fails to identify major spectral peaks or substantially selects irrelevant peaks as important evidence
	m/z annotation accuracy	15	Evaluate consistency between reported precursor/fragment m/z values and the experimental spectrum	5: Key precursor and fragment m/z values are accurately annotated with ppm error ≤ 5 ppm; 3: Major m/z values are generally reliable with 5 ppm $<$ ppm error ≤ 20 ppm; 1: Incorrect or mismatched m/z annotation, or ppm error > 20 ppm
Key Fragment	Fragment completeness	15	Evaluate the coverage of major diagnostic fragment ions and their relevant fragment-level information	5: Covers the major diagnostic fragments and provides meaningful fragment-level information, including relevant fragment relationships and mass differences; 3: Reports several major fragments but provides incomplete coverage or limited interpretation of fragment relationships; 1: Major diagnostic fragments are substantially absent, incorrectly reported, or poorly described
	Neutral loss validity	20	Evaluate the accuracy of neutral-loss mass-difference calculations derived from precursor-fragment or fragment-fragment relationships	5: All identifiable neutral-loss values are correctly calculated; 3: At least one neutral-loss value is correctly calculated; 1: No neutral-loss values are correctly calculated, or reported values are incorrect or marked as unknown
Intensity	Precursor annotation	10	Evaluate the accuracy of reported precursor metadata and, where supported, precursor ion-form interpretation	5: Correctly extracts precursor metadata and additionally provides reasonable ion/adduct hypotheses with explicit uncertainty statements when supported by the spectral evidence; 3: Correctly extracts and reports known precursor m/z and charge state; 1: Fails to correctly extract or report available precursor metadata, including precursor m/z or charge state
	Relative intensity	5	Evaluate the extraction and interpretation of peak abundance information	5: Correctly identifies dominant fragments/base peak, reports relative-intensity values, and further uses intensity distribution to evaluate fragment importance; 3: Successfully extracts and reports base-peak and major-peak relative-intensity numerical values; 1: Cannot identify or report dominant peaks, base peak, or relative-intensity information
	Sorting logic	8	Evaluate the analytical organization and prioritization of reported spectral peaks	5: Organizes fragments according to analytical importance by integrating intensity, diagnostic value, and fragmentation relevance; 3: Organizes fragments according to simple criteria such as intensity or m/z ; 1: Lists peaks without meaningful analytical organization or prioritization
Chemical Interpretation	Hypothesis rationality	10	Evaluate the chemical plausibility and evidence support of fragment-assignment or structural hypotheses	5: Provides evidence-supported fragment-assignment or structural hypotheses, evaluates their chemical plausibility, and explicitly states uncertainty where the interpretation is not uniquely determined; 3: Provides reasonable fragment-assignment or structural hypotheses based on the observed MS/MS evidence but with limited chemical justification; 1: Fails to provide meaningful fragment-assignment or structural hypotheses, or provides unsupported and chemically implausible interpretations
	Mass verification	7	Verify consistency among precursor-fragment and fragment-fragment mass relationships	5: Correctly extracts and calculates mass relationships, systematically verifies precursor-fragment and fragment-fragment consistency, and explicitly discusses potential uncertainties or alternative interpretations; 3: Correctly extracts precursor and fragment m/z values and calculates basic mass differences, including neutral losses, but provides limited verification of overall mass consistency; 1: Fails to correctly extract precursor or fragment m/z values required for mass relationship analysis, or reports incorrect mass differences

Note: All items adopt ordinal three-level scoring ($\{5, 3, 1\}$). The final weighted score for each item is calculated as raw score $\times$ (Weight / 5). The total score ranges from 20 to 100 under this scoring scheme.

Table S8: Comprehensive scoring criteria for structure report generation (full score: 100)

Dimension Category	Scoring Item	Weight (%)	Scoring Criteria
Molecular Scaffold	Scaffold integrity	10	5: Complete scaffold identification with accurate structural features; 3: Major scaffold correctly identified with minor structural incompleteness; 1: Core scaffold incorrectly recognized or missing.
	Substituent identification and positioning	10	5: All major and minor substituents and positions correctly identified; 3: Major substituents correctly identified with minor positional uncertainty; 1: Key substituent assignment or positioning errors.
Functional Groups	Functional group completeness	15	5: All major functional groups correctly identified without critical false assignments; 3: Major groups identified with minor omissions or misclassification of secondary groups; 1: Major functional groups missing or incorrectly assigned.
	Chemical property annotation	10	5: Chemical properties including heteroatom features, acid-base properties, and adduct compatibility are accurately characterized; 3: Major chemical properties correctly identified with minor omissions; 1: Fundamental chemical property assignments are incorrect.
Chemical Bonds	Fragmentation-related bond annotation	15	5: Key fragmentation-prone bonds and associated neutral losses are correctly identified; 3: Major fragmentation patterns captured with minor missing assignments; 1: Key fragmentation sites or neutral losses incorrectly predicted.
	Ring system characterization	10	5: All ring systems correctly identified and structurally characterized; 3: Major ring systems recognized with incomplete classification; 1: Core ring systems incorrectly identified or missing.
Physicochemical Properties	Formula accuracy	15	5: Molecular formula completely matches the elemental composition derived from SMILES; 1: Molecular formula is incorrect or unavailable, including incorrect elemental composition or unknown output.
	Exact mass accuracy	15	5: Exact mass matches the theoretical value ($\text{ppm error} \leq 1 \text{ ppm}$); 3: Exact mass shows minor deviation from the theoretical value ($1 \text{ ppm} < \text{error} \leq 10 \text{ ppm}$); 1: Exact mass has significant deviation ($\text{ppm error} > 10 \text{ ppm}$) or is unavailable (unknown output).

Note: Most items adopt ordinal three-level scoring ($\{5, 3, 1\}$). *Formula accuracy* uses two-level scoring ($\{5, 1\}$). The final weighted score for each item is calculated as $\text{raw score} \times (\text{Weight} / 5)$.

Appendix C: CASMI 2022 Benchmark Evaluation and Reproducibility

C.1 CASMI 2022 Benchmark and Candidate Space

The CASMI 2022 benchmark was used as the primary evaluation set for end-to-end MS²-to-structure identification. The benchmark contains 500 test samples spanning multiple ionization modes and adduct forms. Per-adduct sample statistics and corresponding retrieval performance are summarized in Table S9, and the complete set of 500 samples was retained for evaluation without additional subsampling. To ensure a controlled comparison across methods, the candidate retrieval space was fixed to the 497 unique SMILES entries obtained after canonicalization and deduplication of the 500 benchmark compounds. Thus, every method was evaluated against an identical candidate set of 497 structures for each query spectrum. No precursor mass filtering (e.g., 10 ppm or 0.001 Da tolerance) or other candidate-space reduction was applied. No method was provided with an expanded external candidate database during evaluation. MS²Agent achieves an overall Top-1 accuracy of 0.784 on the complete 500-sample CASMI 2022 test set.

The candidate structures were standardized and canonicalized before evaluation. Molecular representations used by the evaluation pipeline were generated using the same structure-standardization procedure described in Appendix A. Candidate identifiers were retained throughout the retrieval and reranking procedures to enable unambiguous matching between predictions and reference structures.

The benchmark-level evaluation therefore differs from methods that evaluate only selected ion forms or a subset of CASMI entries. In the present study, all 500 test samples were evaluated under a unified candidate space while preserving the ionization and adduct information associated with each experimental spectrum.

Table S9: Stratified identification performance of MS²Agent on CASMI 2022 by adduct type.

Adduct	<i>N</i>	Recall@1	Recall@3	Recall@5	Recall@10	MRR
<i>Positive-ion (304)</i>						
[M+H] ⁺	177	0.825	0.836	0.836	0.842	0.8382
[M+NH ₄] ⁺	67	0.746	0.776	0.776	0.791	0.7736
[M+Na] ⁺	60	0.783	0.800	0.800	0.800	0.7989
<i>Negative-ion (196)</i>						
[M-H] ⁻	108	0.796	0.806	0.806	0.824	0.8119
[M+FA-H] ⁻	67	0.701	0.731	0.731	0.731	0.7242
[M+Cl] ⁻	21	0.762	0.762	0.762	0.762	0.7735
Full Pool	500	0.784	0.800	0.800	0.808	0.8012

C.2 ChemEmbed Evaluation

ChemEmbed was evaluated as a conventional embedding-based retrieval baseline. The publicly available ChemEmbed model and its corresponding embedding-based retrieval procedure were used without task-specific retraining.

Because the original ChemEmbed study reported results only on a subset of the CASMI 2022 benchmark—107 positive-ion and 64 negative-ion samples—we evaluated ChemEmbed on exactly these two subsets to enable a direct and fair comparison with the published results. The same 107 and 64 samples were also evaluated under MS²Agent and all other methods using the identical 497 candidate SMILES entries, ensuring that every method was compared on the same queries and the same candidate space.

For each query spectrum, the corresponding spectrum representation was compared with the representations of the 497 candidate SMILES entries in the fixed benchmark candidate space. Candidate structures were ranked according to embedding similarity or distance, and Recall@*K* was calculated from the resulting ranking.

Cosine similarity, Euclidean distance, and a best-of-two fusion (which takes, for each candidate, the better rank from the cosine and Euclidean lists) were all evaluated to characterize the effect of the retrieval metric. The best-of-two result was used as the primary ChemEmbed value in the main-text benchmark comparison because it provided the highest

overall retrieval accuracy among the three settings.

Table S10: Comparison of MS²Agent and ChemEmbed on the original ChemEmbed evaluation subsets of CASMI 2022.

Positive-ion subset ($n = 107$)				
Method	Top-1	Top-3	Top-5	Top-10
MS ² Agent	0.757	0.766	0.766	0.776
ChemEmbed-Cosine	0.1308	0.1869	0.2430	0.3738
ChemEmbed-Euclidean	0.1963	0.3458	0.4206	0.5234
ChemEmbed-Best-of-two	0.2150	0.3738	0.4486	0.5607
Negative-ion subset ($n = 64$)				
Method	Top-1	Top-3	Top-5	Top-10
MS ² Agent	0.844	0.844	0.844	0.844
ChemEmbed-Cosine	0.0625	0.1094	0.1719	0.2500
ChemEmbed-Euclidean	0.0781	0.1563	0.2188	0.2969
ChemEmbed-Best-of-two	0.1250	0.2188	0.2969	0.3750

Note. Results are reported on the 107 positive-ion and 64 negative-ion evaluation subsets from the original ChemEmbed study. Cosine and Euclidean denote retrieval based on a single distance metric; Best-of-two takes, for each candidate, the better rank from the cosine and Euclidean lists. All methods were evaluated using the same 497 unique candidate SMILES entries.

C.3 CSU-MS² Evaluation

CSU-MS² was evaluated using the authors’ publicly released implementation (<https://github.com/tingxiecsu/CSU-MS2>) without retraining on the CASMI 2022 test set. The candidate retrieval space was restricted to the same 497 unique candidate SMILES entries used for all other methods.

To ensure reproducibility, we followed the inference procedure implemented in the official CSU-MS² repository, particularly the library-search workflow provided in `search_library.py` and `search_user_defined_library.py`. No additional model training or method-specific optimization was introduced for CSU-MS².

C.4 General-Purpose Agent Baseline: Codex

To assess whether general-purpose agentic reasoning alone is sufficient for MS²-to-structure identification, Codex combined with GPT-5.4-mini was included as a general-purpose agentic

baseline. The evaluation was performed on the complete CASMI 2022 test set under the same unified benchmark setting used for all other methods.

For each test sample, Codex was provided with the corresponding MS² spectrum information and precursor information, together with the predefined candidate library containing the same 497 unique SMILES entries used throughout the evaluation. The ground-truth structure was not provided to the agent. Codex was instructed to analyze the observed spectral information, compare the spectrum with the candidate structures, and return a ranked list of candidate structures for the given query spectrum.

No task-specific training, fine-tuning, or parameter optimization was performed for Codex. The baseline therefore evaluates the capability of a general-purpose agent, equipped with GPT-5.4-mini, to perform MS²-to-structure identification through general-purpose reasoning rather than through a specialized mass-spectrometry representation or retrieval model.

To ensure a controlled comparison, the candidate library was fixed to the same 497 unique SMILES entries used by the specialized retrieval methods. The same test samples were evaluated for all methods, and candidate rankings were converted into Recall@1, Recall@3, Recall@5, and Recall@10 using the unified evaluation script. Candidates outside the predefined benchmark library were not considered when calculating retrieval performance.

The Codex baseline achieved Recall@1/3/5/10 values of 0.156/0.220/0.276/0.350, respectively, substantially below the performance of the task-specific MS²Agent framework using the same GPT-5.4-mini backbone. This result indicates that general-purpose agentic reasoning alone provides limited domain-specific spectral discrimination for fine-grained MS²-structure matching. The Codex baseline therefore provides a controlled reference for evaluating the contribution of the task-specific reasoning, spectral analysis, and candidate-selection mechanisms implemented in MS²Agent, and supports the value of domain-specialized modules for metabolite identification from tandem mass spectra.

C.5 CASMI Evaluation Pipeline

All methods were evaluated using the same 497 unique candidate SMILES entries without additional candidate-space filtering. The test samples and evaluation criteria were fixed across all methods, while each method followed its respective inference and candidate-ranking procedure.

C.5.1 Evaluation Metrics

All method-specific rankings were converted into Recall@1, Recall@3, Recall@5, Recall@10, and Mean Reciprocal Rank (MRR) using a unified evaluation script.

Recall@K. For a benchmark containing N test samples, Recall@ K was calculated as

$$\text{Recall@}K = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(r_i \leq K), \quad (\text{S14})$$

where r_i denotes the rank of the ground-truth structure for the i -th test sample and $\mathbb{I}(\cdot)$ is the indicator function. A prediction was counted as correct when the ground-truth standardized structure appeared within the top- K ranked candidates. Structure identity was determined using canonical SMILES generated by the benchmark structure-standardization procedure. Candidates outside the predefined benchmark candidate library were excluded from metric calculation.

Mean Reciprocal Rank (MRR). The average of the reciprocal ranks of the reference structure across all queries:

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i}, \quad (\text{S15})$$

where N is the number of queries and rank_i is the position of the reference structure in the candidate list for query i . If the reference structure is not found among the candidates, $1/\text{rank}_i$ is taken as zero.

C.5.2 Robustness Evaluation Supplementary Results

This appendix presents the detailed quantitative results of the robustness evaluation for MS²Rep-0.8B-GRPO on the CASMI 2022 dataset under three typical experimental interference scenarios: noise perturbation, collision energy variation, and spectral degradation. Under all interference conditions, the same 497 candidate SMILES entries were used without additional candidate-space filtering.

Table S11: Robustness performance of MS²Rep-0.8B-GRPO under interference conditions on CASMI 2022

Interference Condition	Recall@1	Recall@3	Recall@5	Recall@10
Baseline	0.784	0.800	0.800	0.808
Noise Perturbation (SNR)				
SNR5	0.770	0.802	0.804	0.812
SNR10	0.758	0.802	0.806	0.810
SNR20	0.682	0.714	0.718	0.718
Collision Energy (eV)				
15 eV	0.780	0.806	0.808	0.808
35 eV	0.766	0.802	0.806	0.806
55 eV	0.780	0.804	0.808	0.816
Spectral Degradation				
Light	0.770	0.804	0.804	0.808
Medium	0.784	0.806	0.814	0.816
Heavy	0.792	0.814	0.816	0.818

Appendix D: MassSpecGym v1.5 Evaluation Protocol

This appendix documents the dataset preparation, candidate retrieval, and evaluation procedures used for the MassSpecGym v1.5 benchmark results.

D.1 Dataset Preparation and Train–Test Separation

MS²Agent was trained on 12,000 molecules randomly sampled from the in-house database of approximately 1.6 million MS/MS–SMILES associations described in Section A.2. This training subset is independent of the official MassSpecGym training split. To prevent data leakage, test queries were filtered against the 12,000 training molecules in two stages:

1. **InChIKey-14 deduplication:** test queries whose 14-character InChIKey prefix matched any molecule in the training set were removed, eliminating exact connectivity overlaps.
2. **MCES filtering:** remaining test queries whose maximum common edge subgraph (MCES) distance to any training molecule was less than 10 were further excluded. The MCES distance threshold of 10 follows the structural separation criterion used in the original MassSpecGym benchmark, ensuring that no test molecule is structurally near-duplicate of a training molecule.

After the two-stage train–test filtering described above, 4,295 query spectra covering 967 unique SMILES (940 unique InChIKey-14 identifiers) were retained for evaluation.

Evaluation Metrics

Evaluation metrics are defined in Appendix C (Recall@ k and Mean Reciprocal Rank). All metrics reported in this section follow the same definitions and evaluation protocol.

D.2 Baseline Reproduction

JESTR and MIST were evaluated on the same 4,295-query subset under identical candidate retrieval and metric computation procedures. We utilized the publicly released implementations and pre-trained weights from the respective authors¹. Because the original publications report only full-set aggregate metrics, direct comparison requires reproduction on the shared subset. JESTR was run in two configurations: *Mass* mode (spectral ranking without formula information) and *Formula* mode (ranking constrained to candidates sharing the reference molecular formula). MIST was similarly evaluated in Mass mode (using its internally predicted molecular formula) and Formula mode (using the reference formula).

All baseline results were obtained through the MassSpecGym benchmark system, which reports aggregate Recall@1, Recall@5, and Recall@20.

¹JESTR: <https://github.com/HassounLab/JESTR1>; MIST: <https://github.com/samgoldman97/mist>

Appendix E: Isomer Discrimination Supplementary Results

All candidate structures were retrieved from PubChem (accessed August 2026; PubChem is continuously updated and lacks static versioning) via molecular formula matching. For each query, the candidate pool comprised all PubChem entries sharing the same molecular formula, yielding pool sizes ranging from 99 to 14,178 structures. The initial matcher retrieved the top-200 candidates from the full pool, and a reranker model subsequently reordered these 200 candidates to produce the final ranked candidate list.

E.1 Baseline Reproduction

JESTR and MIST were benchmarked on the identical 324-query isomer subset using unified candidate retrieval pipelines and metric calculation protocols. We adopted the publicly available codebases and pre-trained weights released by the original authors (see Appendix D for repository links).

E.2 Successful Discrimination: Saponarin and Its Glycosylation Isomers

The successful case corresponded to saponarin (apigenin 6-C-glucoside-7-O-glucoside; $C_{27}H_{30}O_{15}$), which was evaluated against a candidate pool of 269 structures. The maximum ECFP4 Tanimoto similarity between the ground truth and the competing candidates was $S_{\max} = 0.770$. MS²Agent correctly ranked saponarin at Top 1, distinguishing it from closely related flavonoid glycosides differing in glycosylation position, C- versus O-glycosidic linkage, sugar configuration, and hydroxyl substitution pattern.

The most structurally similar competing candidate was ranked fifth (Tanimoto = 0.770). This candidate retained a related flavonoid scaffold and two sugar moieties but exhibited an

Table S12: Overview of molecular formulas, SMILES counts, spectra counts, PubChem candidate pool sizes, and CE ranges. Candidate pool sizes were obtained from PubChem via exact molecular formula matching.

Formula	#SMILES	#Query	PubChem Pool	CE range (eV)
C27H30O15	10	31	269	6–20
C26H28O14	10	28	146	6–10
C41H74NO8P	10	25	210	30–30
C20H34O5	7	25	794	6–30
C44H78NO9P	10	22	99	30–30
C43H76NO9P	10	21	150	30–30
C43H78NO9P	10	21	119	30–30
C21H24N2O3	8	20	13,870	6–30
C7H5NO4	4	13	244	15–40
C5H11NO2	4	10	1,431	10–40
C20H32O4	5	10	2,489	10–40
C15H22O3	10	10	199	10–200
C6H15NO	3	9	778	10–40
C10H8O4	5	9	1,142	6–30
C10H10O3	4	8	2,754	6–50
C8H8O2	3	8	909	20–40
C9H8O4	5	7	1,027	6–30
C24H40O3	2	6	1,130	10–40
C11H13NO4	3	5	6,657	30–80
C14H12N2	2	4	2,216	6–80
C11H16N2O	1	4	14,178	30–60
C7H10O4	1	3	1,755	20–60
C10H9NO3	1	3	4,043	17–40
C13H16N2	1	3	6,752	20–20
C12H8N2O	2	3	917	10–40
C9H9N	1	3	654	35–35
C8H13NO2	1	3	6,928	30–120
C14H22O	2	2	5,645	15–50
C10H15NO	1	1	7,772	6–60
C10H13NO	1	1	6,323	30–90
C18H18O5	1	1	2,880	20–50
C11H17N	1	1	4,343	20–45
C12H11NO	1	1	2,828	40–120
C10H12N2O2	1	1	9,549	15–180
C7H10O5	1	1	1,073	10–30
C13H11NO2	1	1	3,189	30–75
Total	143	324	99–14,178	

alternative 8-glycosyl substitution pattern rather than the 6-C/7-O glycosylation pattern of saponarin. The successful ranking indicates that, for this query, the experimental spectrum contained sufficient information for the framework to prioritize the correct combination of aglycone substitution, glycosylation position, and linkage type. Table S13 lists the Top-10 ranked candidates.

Table S13: Top-10 candidate ranking for saponarin (successful case)

Rank	Tanimoto	Key Difference from Ground Truth
1	1.000	Ground truth —saponarin (apigenin 6-C-glucoside-7-O-glucoside)
2	0.461	Flavonoid glycoside with a different glycosylation position
3	0.580	7-Glycosylated flavone with a different B-ring hydroxylation pattern
4	0.609	5,7-Dihydroxyflavone glycoside with an alternative substitution pattern
5	0.770	8-Glycosyl-substituted positional isomer
6	0.758	5,7-Dihydroxy-6-glycosylated flavonoid glycoside
7	0.514	3',4'-Dihydroxyflavonoid glycoside with a different glycosylation pattern
8	0.719	Flavonoid glycoside with an alternative sugar arrangement
9	0.465	3,7-Diglycosylated flavonoid positional isomer
10	0.420	Flavonoid glycoside with an alternative sugar and substitution pattern

E.3 Failure Case: Kaempferol 3-O-Neohesperidoside and Related Glycosides

The failure case corresponded to kaempferol 3-O-neohesperidoside ($C_{27}H_{30}O_{15}$), evaluated against the same candidate pool of 269 structures. In this compound, a neohesperidose moiety consisting of rhamnose and glucose linked through an α -(1 $\rightarrow$ 2) glycosidic bond is attached to the C-3 position of the kaempferol scaffold. The candidate space contained a competing structure with an ECFP4 similarity of $S_{\max} = 1.000$, indicating a highly ambiguous structural retrieval setting.

The ground-truth structure was ranked 14th, with 13 incorrect candidates placed above it. The Top 1 candidate retained a related flavonoid-disaccharide architecture but differed

from the ground truth in glycosylation site (Tanimoto = 0.667). Among the 13 candidates ranked above the ground truth, the highest structural similarity was observed for Rank 9 (Tanimoto = 0.892). However, several less similar candidates received higher rankings, further demonstrating that the final ranking was not a direct or monotonic mapping of ECFP4 similarity.

This case illustrates the current resolution boundary of MS²Agent. Although the framework integrates spectral and structural evidence to rerank candidates, it cannot reliably distinguish fine-grained differences in glycosylation site, interglycosidic connectivity, sugar configuration, or aglycone substitution when these features are weakly or ambiguously represented in the experimental fragmentation pattern. Under such conditions, the output should be interpreted as evidence-guided candidate prioritization rather than definitive structure determination. Table S14 presents the candidate rankings through the ground-truth position.

Table S14: Candidate ranking through the ground-truth position for kaempferol 3-O-neohesperidoside (failure case)

Rank	Tanimoto	Key Difference from Ground Truth
1	0.667	Alternative C-8 glycosylation pattern
2	0.512	Alternative C-7 glycosylation pattern
3	0.420	Different sugar-ring structure and glycosylation arrangement
4	0.841	Rutin-like disaccharide connectivity isomer
5	0.440	Different hydroxyl substitution pattern on the flavonoid B-ring
6	0.761	Glycosidic linkage-position isomer
7	0.694	Different hydroxyl substitution and sugar arrangement
8	0.703	Alternative sugar-ring and glycosidic linkage configuration
9	0.892	Closely related rutin-like glycoside with altered hydroxyl substitution
10	0.597	Alternative sugar and glycosylation pattern
11	0.627	Alternative sugar and linkage arrangement
12	0.556	Alternative sugar-ring structure and glycosylation pattern
13	0.500	Alternative sugar configuration and attachment pattern
14	1.000	Ground truth —kaempferol 3-O-neohesperidoside

Appendix F: Experimental Validation on *Angelica pubescens*

Standard Compounds

This appendix describes the experimental acquisition and MS² identification of 20 structurally characterized standard compounds obtained from *Angelica pubescens*. This experiment was conducted to evaluate MS²Agent on experimentally acquired spectra from authentic standard compounds. Detailed sample preparation, UPLC conditions, and mass spectrometric parameters are provided below. All 20 standard compound datasets and full interpretation reports are provided in the Supplementary Materials.

F.1 Standard Compounds

Twenty structurally characterized standard compounds associated with *Angelica pubescens* were selected for experimental validation. The compound identities and reference information are summarized in Table S15.

Table S15: Reference standard compounds used for experimental validation.

No.	Compound	Formula	Exact mass
1	Isochlorogenic acid A	C ₂₅ H ₂₄ O ₁₂	516.1268
2	Neochlorogenic acid	C ₁₆ H ₁₈ O ₉	354.0951
3	Isochlorogenic acid B	C ₂₅ H ₂₄ O ₁₂	516.1268
4	Oxypeucedanin	C ₁₆ H ₁₄ O ₅	286.0841
5	Isochlorogenic acid C	C ₂₅ H ₂₄ O ₁₂	516.1268
6	Phellopterin	C ₁₇ H ₁₆ O ₅	300.0998
7	Imperatorin	C ₁₆ H ₁₄ O ₄	270.0892
8	Xanthotoxol	C ₁₁ H ₆ O ₄	202.0266
9	Isopsoralen	C ₁₁ H ₆ O ₃	186.0317
10	Nodakenin	C ₂₀ H ₂₄ O ₉	408.1420
11	Chlorogenic acid	C ₁₆ H ₁₈ O ₉	354.0951
12	Hydrated oxypeucedanin	C ₁₆ H ₁₆ O ₆	304.0947
13	Osthole	C ₁₅ H ₁₆ O ₃	244.1099
14	Caffeic acid	C ₉ H ₈ O ₄	180.0423
15	Gallic acid	C ₇ H ₆ O ₅	170.0215
16	Bergaptol	C ₁₁ H ₆ O ₄	202.0266
17	Psoralen	C ₁₁ H ₆ O ₃	186.0317
18	Xanthotoxin	C ₁₂ H ₈ O ₄	216.0423
19	Isoferulic acid	C ₁₀ H ₁₀ O ₄	194.0579
20	Bergapten	C ₁₂ H ₈ O ₄	216.0423

F.2 Sample Preparation

Angelica pubescens Radix decoction pieces were prepared according to the 2020 edition of the Chinese Pharmacopoeia. For each sample, 0.1 g of the decoction pieces was accurately weighed, placed in a 10 mL volumetric flask, and extracted with 10 mL of mixed solvent (methanol:water, 1:1, v/v). After ultrasonic extraction for 30 min, the lost solvent was replenished with the same mixed solvent. The solution was then filtered through a 0.22 μm microporous membrane, and the filtrate was collected for subsequent UPLC–MS analysis.

All samples were prepared using the same procedure to minimize variation introduced by sample preparation.

F.3 UPLC Conditions

Chromatographic separation was performed using an Agilent 1290 Infinity UPLC system equipped with a binary solvent delivery pump, online microdegasser, thermostatted autosampler, and column oven. Separation was achieved using a Thermo Fisher Scientific Hypersil GOLD C18 column (100 $\times$ 2.1 mm, 1.9 μm).

Table S16: UPLC chromatographic conditions used for experimental MS² acquisition.

Parameter	Condition
Instrument	Agilent 1290 Infinity
Column	Hypersil GOLD C18, 100 $\times$ 2.1 mm
Particle size	1.9 μm
Column temperature	35°C
Mobile phase A	0.1% (v/v) formic acid in water
Mobile phase B	Acetonitrile
Flow rate	0.3 mL/min
Injection volume	2 μL
Gradient program	See Table S17
Total run time	30 min

Table S17: Gradient elution program for UPLC separation.

Time (min)	A (%)	B (%)
0.00	97	3
5.00	78	22
15.00	58	42
20.00	40	60
25.00	0	100
28.00	0	100
28.01	97	3
30.00	97	3

F.4 Mass Spectrometric Conditions

Mass spectrometric data were acquired using an Agilent 6545A quadrupole time-of-flight (Q-TOF) mass spectrometer equipped with an Agilent Jet Stream electrospray ionization (ESI) source.

Table S18: Mass spectrometric conditions used for experimental MS² acquisition.

Parameter	Condition
Mass spectrometer	Agilent 6545A Q-TOF
Ionization source	Agilent Jet Stream ESI
Ionization mode	Positive / Negative
Scan range	m/z 50–1700
MS acquisition mode	Auto MS/MS
MS ² acquisition mode	Top 3 precursors per cycle
Collision energy	20 eV
Capillary voltage	3500 V (positive) / 3000 V (negative)
Fragmentor voltage	130 V
Drying gas (N ₂) flow	8 L/min
Gas temperature	320°C
Nebulizer pressure	35 psi

F.5 MS² Spectrum Processing

The experimentally acquired MS² spectra were converted into the standardized spectral representation required by MS²Agent.

The following processing steps were applied:

1. Extract the precursor ion and corresponding MS² spectrum from the raw acquisition data.
2. Record the precursor m/z , ion polarity, and precursor ion form.
3. Extract the fragment-ion m/z and intensity pairs.
4. Apply the predefined peak filtering and intensity normalization procedure.
5. Convert the processed spectrum into the input representation required by MS²Agent.

F.6 MS²Agent Identification Procedure

The experimentally acquired spectra were analyzed using MS²Agent. The identification procedure consisted of three major stages: candidate generation, candidate-level spectral and structural reasoning, and final candidate ranking. Two complementary candidate retrieval routes were employed.

F.6.1 Candidate Generation

Route 1: PubChem-based retrieval. For each spectrum, the MS report generation agent first inferred up to 10 candidate molecular formulas from the precursor m/z , neutral losses, and fragment-ion patterns. For each candidate formula, up to 200 structurally diverse compounds were retrieved from the PubChem database (accessed May 2026; PubChem is continuously updated and lacks static versioning) within a mass tolerance of 10 ppm, yielding up to approximately 2000 candidate structures in total. These candidates were then encoded using Qwen3-Embedding-0.8B and passed through the BERT-Sim classifier, which selected the top-40 most relevant candidates for subsequent LLM-based reranking.

Route 2: Self-built library matching. In parallel, a self-built compound library of 76 *Angelica pubescens*-related compounds was used as a targeted candidate source. The observed MS² spectrum was encoded via Qwen3-Embedding-0.8B, and cosine similarity was computed against all 76 library entries. The top-20 candidates were selected for LLM-based reranking.

F.6.2 Spectral and Structural Reasoning

For each candidate set, MS²Agent evaluated the consistency between the observed MS² spectrum and each proposed molecular structure. The reasoning process considered diagnostic fragment ions, neutral-loss relationships, precursor-ion consistency, and structural plausibility.

The agent configuration used for the experimental validation was as follows:

- **MS report generation model (Agent 1):** MS2Rep-GRPO-0.8B
- **Structure report generation model (Agent 2):** MS2Rep-GRPO-0.8B
- **Embedding model (Agent 3):** Qwen3-Embedding-0.8B or Bert-sim
- **LLM reranker model (Agent 4):** GPT-5.4-mini
- **Number of candidates reranked (PubChem route):** 40
- **Number of candidates reranked (self-built route):** 20

F.6.3 Candidate Ranking

Candidate structures were ranked by the LLM reranker under the five-level priority criterion: (1) exact mass and adduct consistency; (2) neutral loss and representative fragment consistency; (3) acid–base properties, heteroatom composition, and functional group compatibility; (4) fragile bond and likely loss consistency; (5) broad scaffold or chemical family similarity. The highest-ranked candidate was taken as the predicted molecular identity.

No manual modification of the final ranking was performed.

F.7 Identification Results for the 20 Standard Compounds

The identification results for the 20 structurally characterized standard compounds under both candidate retrieval routes are summarized in Table S19. Detailed per-compound identification results are provided in the Supplementary Materials and will be made publicly available in the GitHub.

F.8 Dimension-wise Scoring of Osthole Identification Reports

To evaluate the quality of MS²Agent’s structural reasoning for osthole, the generated MS report and SMILES report were independently scored by domain experts using the predefined

Table S19: Summary of experimental validation results under both candidate retrieval routes.

Metric	PubChem route	Self-built library
Number of reference standards	20	20
Number of spectra successfully acquired	20	20
Recall@1	0.65	0.90
Recall@5	0.85	0.95
Recall@10	0.95	1.00
Recall@20	0.95	1.00

expert scoring criteria (full score: 100). The dimension-wise scoring results are presented in Table S20.

Table S20: Dimension-wise scoring results of osthole-related reports based on the predefined expert scoring criteria (full score: 100)

Module	Scoring Item	Weight (%)	Score	Evaluation rationale
MS Report (64.0)	Peak filtering	10	6.0	Major peaks identified, but no systematic discrimination of informative vs. background peaks documented.
	m/z annotation accuracy	15	15.0	Precursor and fragment m/z values matched raw data within 0.4 ppm.
	Fragment completeness	15	9.0	Major fragments reported, but coverage and relationship interpretation limited.
	Neutral loss validity	20	20.0	All three mass differences correctly calculated.
	Precursor annotation	10	2.0	Precursor charge omitted; m/z reported as neutral-mass hypothesis.
	Relative intensity interpretation	5	1.0	Base peak identified, but numerical relative intensities not reported.
	Sorting logic	8	4.8	Ordered by intensity without diagnostic-relevance prioritization.
	Hypothesis rationality	10	2.0	Unsupported sulfur-containing neutral-loss interpretation proposed for the sulfur-free molecule.
	Mass verification	7	4.2	Basic mass differences correct, but overall mass consistency not systematically verified.
SMILES Report (68)	Scaffold integrity	10	8	Core scaffold was correctly identified
	Substituent identification and positioning	10	4	Substituent positions were inaccurate
	Functional group completeness	15	6	Minor functional groups were incompletely described
	Chemical property annotation	10	8	Chemical properties were reasonably annotated
	Fragmentation-prone bond annotation	15	6	Cleavage trends were partially inaccurate
	Ring system characterization	10	6	Ring system was recognized but incompletely analyzed
	Formula accuracy	15	15	Formula matched $C_{15}H_{16}O_3$
	Exact mass accuracy	15	15	Exact mass was accurately calculated

F.9 Raw MS² Spectrum of Osthole (MGF Format)

The representative raw MS² spectrum of osthole acquired in positive ionization mode is presented below in MGF format. The precursor m/z was 245.1175 with a charge state of 1⁺.

```
BEGIN IONS
PEPMASS=245.1174683
SMILES=CC(=CCC1=C(C=CC2=C1OC(=O)C=C2)OC)C
CHARGE=1+
TITLE=BZP-ZMH-POS-SCZS.d, MS/MS of 245.1174683 1+ at 17.1991166666667 mins
RTINSECONDS=1031.947
32.6741486673359 18.125
34.4135092904415 11.25
34.722215352581 13
47.3979578123978 10.125
47.7881961086302 20.025
49.812172004361 10.125
68.3758967752247 34.34722
74.5990551326671 21
76.5107252317784 21.01786
77.0382916558865 488.9141
77.0440904892685 46.025
77.0536550902531 12
77.083996304535 10.125
77.9260273878055 34.375
102.464428398377 68.66666
103.054785722753 931.1278
103.072408162093 13.01786
103.078826617651 28.92308
103.207088199854 41.44643
103.251168349696 32.375
103.270280041198 54.125
103.522506524689 23.375
103.529262779219 13.125
103.690321475701 20
104.334371962099 29.25
104.477954307186 58.92046
104.485129763992 22.5
105.067294477088 210.15
105.138928619384 12.75
112.370693466469 17.16072
```

114.779079051675 60.0625
115.053679382573 128.25
116.208053011211 27.5
118.457322836731 22.5
131.04874963301 4300.089
131.058505691946 121.7206
131.073604477765 118.1635
131.078022664502 31.78125
131.097410766297 12.25
131.108310892865 32.11184
131.111487141325 28.32237
131.11807195968 31.625
131.121326871754 58.125
131.129197742142 26.01786
131.146983487848 15.125
131.158230568698 10.125
131.202678265995 23.0625
131.207021879373 19.125
131.21064959124 30.01389
131.214173603719 23
131.412612357904 33.0625
131.422094117549 11.25
131.465483585912 69.01136
131.474450065387 30
131.506807851103 36.16071
131.565537051871 17.25
131.634771453577 17.025
131.797719832519 55.025
131.806478041211 20.25
131.938515396702 50.33333
132.055965542742 199.125
132.165295315488 42
132.301090558737 22.5
132.305950005304 14.25
132.617798914234 42.0625
132.626924946041 22.01389
133.029231930752 44.875
133.064111327345 270
133.073073545338 47.25
133.087437675258 14
133.120904095419 61.8848

133.27297161568 15
133.492919656586 101.0625
134.067095716566 141.1118
135.825630810262 11.08333
140.276640600116 11
141.314626344798 14.025
159.0425387453 350.15
159.07128203289 29.2
161.060556792709 164.0083
161.387531382438 44.10227
161.395736374604 15.75
162.534607238108 13.08333
179.766096592946 20.16072
181.253730122502 10
189.052622479294 155.5
189.061565327967 13.25
189.33548955254 13.025
201.085900860759 11.125
210.016231268259 13
212.688667821463 14.625
214.460582680926 15
308.377412196814 13.08333
393.967094794824 23.08333
403.208046117547 15.08333
450.794157797338 11.5
462.69827383517 11.25
472.550289064533 12.5
487.616822633476 30.75
509.702285026422 53.04167
509.715865728728 30.25
580.820419387839 10.25
630.753424860162 16.08333
645.129380816318 25.875
657.816508658315 12.5
681.555770873527 47.45
726.199107339562 17
788.795458821841 14.625
875.142278347106 30.625
875.730732983234 18
875.751518855824 11
875.972732093202 28.2

906.551418770662 14.375
 927.593860160272 47.34722
 938.469012097831 26.0625
 946.644313050916 42.75
 1004.51440632439 31
 1105.24039527003 11
 1188.88909748723 19.0625
 1243.63688999563 24.0625
 1249.62792548413 101.0865
 1249.65499870765 19.0625
 1249.88441079067 14
 1274.76754738435 12
 1357.27964151758 13.5
 1362.53958506534 233.9698
 1362.56819182254 70
 1472.58498667309 17
 1475.64734281412 10.125
 END IONS

F.10 MS²Agent Retrieval Signature Report for Osthole

The MS retrieval signature generated by MS²Agent for the osthole spectrum is presented below. The spectrum exhibited high fragmentation richness with dominant fragments at m/z 131.0488, 103.0548, 77.0383, and 159.0425, along with characteristic neutral losses of 114.0687, 142.0627, and 168.0792. A sulfur-containing loss hint was flagged, though osthole does not contain sulfur, which contributed to the retrieval of false-positive candidates in subsequent ranking stages.

```

analysis_type: ms_retrieval_signature
record_id: record_000012
precursor_mz: 245.1175
charge: unknown
ion_mode: positive
likely_ion_forms: unknown
neutral_mass_hypotheses: 245.1175
  
```

peak_count: 129
fragment_peak_count: 129
precursor_dominance: low
fragmentation_richness: high
top_fragments_mz: 131.0488; 103.0548; 77.0383; 159.0425
neutral_losses: 114.0687; 142.0627; 168.0792
chemical_hints: sulfur_containing_loss; aromatic_fragmentation;
 high_fragmentation_richness
retrieval_terms: mz_131_base_peak; mz_103_fragment; neutral_loss_114;
 aromatic_core_pattern; high_fragmentation

F.11 Selected MS² Agent Structure Retrieval Signatures for Osthole Candidates

From the 40 candidates retrieved for the osthole MS² spectrum, we present 10 representative structure retrieval signatures below. The ground truth osthole (record_001776) and diverse confuser scaffolds are included to illustrate the retrieval challenge. The complete set of 40 signatures is provided in the Supplementary Materials.

BEGIN

0

CC(=CCC1=C(C=CC2=C1OC(=O)C=C2)OC)C

analysis_type: structure_retrieval_signature

record_id: record_001776

smiles: CC(=CCC1=C(C=CC2=C1OC(=O)C=C2)OC)C

exact_mass: 244.1099

molecular_formula: C₁₅H₁₆O₃

heteroatom_signature: N0 O3 S0 P0 hal0
hbd: 0
hba: 3
xlogp_class: high
adduct_compatibility: [M+H]⁺; [M+Na]⁺; [M-H]⁻
consistency_check: consistent
scaffold: chromone
ring_system: fused_aromatic
functional_groups: phenol; enol; alkene; alkane
acid_base_class: weak_acid
shape_class: planar_aromatic
likely_losses: CH₃; CH₂; C₂H₄; C₃H₆
fragile_bonds: enol; phenol
retrieval_terms: chromone; phenol; enol; alkene; alkane; weak_acid
END
BEGIN
18
CCCCC1=C(C2=CC=CC=C2C(=O)C1=O)O

analysis_type: structure_retrieval_signature
record_id: record_001794
smiles: CCCCC1=C(C2=CC=CC=C2C(=O)C1=O)O
exact_mass: 244.1099
molecular_formula: C₁₅H₁₆O₃
heteroatom_signature: N0 O3 S0 P0 hal0
hbd: 1
hba: 3

```

xlogp_class: high
adduct_compatibility: [M+H]+; [M+Na]+; [M-H]-
consistency_check: consistent
scaffold: quinone
ring_system: fused_aromatic
functional_groups: quinone; hydroxyl; alkyl_chain
acid_base_class: neutral
shape_class: fused_aromatic
likely_losses: H2O; CO2
fragile_bonds: C-OH; C=O
retrieval_terms: quinone; hydroxyquinone; naphthoquinone; alkyl_substituent;
    C15H16O3; exact_mass_244
END
BEGIN
28
COC(=O)C1CC2=C(C1=O)C=C3CCCCC3=C2

analysis_type: structure_retrieval_signature
record_id: record_001804
smiles: COC(=O)C1CC2=C(C1=O)C=C3CCCCC3=C2
exact_mass: 244.1099
molecular_formula: C15H16O3
heteroatom_signature: N0 O3 S0 P0 hal0
hbd: 0
hba: 3
xlogp_class: high
adduct_compatibility: [M+H]+; [M+Na]+; [M-H]-

```

```

consistency_check: consistent
scaffold: tetralone
ring_system: fused_aromatic
functional_groups: ketone; ester
acid_base_class: neutral
shape_class: fused_aromatic
likely_losses: CO2; CH3OH
fragile_bonds: ester
retrieval_terms: tetralone; methyl ester; ketone; fused aromatic; C15H16O3;
    244.11
END
BEGIN
89
CCC[C@@H](C(=O)O)OC1=CC2=CC=CC=C2C=C1

analysis_type: structure_retrieval_signature
record_id: record_001865
smiles: CCC[C@@H](C(=O)O)OC1=CC2=CC=CC=C2C=C1
exact_mass: 244.1099
molecular_formula: C15H16O3
heteroatom_signature: N0 O3 S0 P0 hal0
hbd: 1
hba: 3
xlogp_class: high
adduct_compatibility: [M-H]-; [M+Cl]-; [M+HCOOH-H]-; [M+CH3COOH-H]-
consistency_check: consistent
scaffold: naphthyl ether

```

```

ring_system: fused_aromatic
functional_groups: carboxylic_acid; ether; alkane
acid_base_class: weak_acid
shape_class: mixed
likely_losses: CO2; H2O; CH3COOH
fragile_bonds: ester; ether
retrieval_terms: naphthyl ether; carboxylic acid; propyl ether; C15H16O3;
    weak acid; CO2 loss
END
BEGIN
118
C=C1CC[C@@H]2[C@@H]([C@@H]3[C@H]1CC=C3C=O)OC(=O)C2=C

analysis_type: structure_retrieval_signature
record_id: record_001894
smiles: C=C1CC[C@@H]2[C@@H]([C@@H]3[C@H]1CC=C3C=O)OC(=O)C2=C
exact_mass: 244.1099
molecular_formula: C15H16O3
heteroatom_signature: O3
hbd: 0
hba: 3
xlogp_class: medium
adduct_compatibility: [M+H]+; [M+Na]+; [M-H]-
consistency_check: consistent
scaffold: tricyclic_alkene
ring_system: fused_aromatic
functional_groups: alpha,beta-unsaturated_ester; exocyclic_alkene; aldehyde;

```

cyclopentene
 acid_base_class: neutral
 shape_class: mixed
 likely_losses: CO2; H2O; CHO
 fragile_bonds: alpha,beta-unsaturated_ester; aldehyde
 retrieval_terms: tricyclic_alkene; alpha,beta-unsaturated_ester; aldehyde;
 exocyclic_alkene; C15H16O3; exact_mass_244
 END
 BEGIN
 142
 COC1=CC(=CC(=C1OC)OC)C2=CC=CC=C2

 analysis_type: structure_retrieval_signature
 record_id: record_001918
 smiles: COC1=CC(=CC(=C1OC)OC)C2=CC=CC=C2
 exact_mass: 244.1099
 molecular_formula: C15H16O3
 heteroatom_signature: N0 O3 S0 P0 hal0
 hbd: 0
 hba: 3
 xlogp_class: high
 adduct_compatibility: [M+H]+; [M+Na]+; [M-H]-
 consistency_check: consistent
 scaffold: dimethoxybenzene
 ring_system: fused_aromatic
 functional_groups: methoxy; phenyl
 acid_base_class: neutral

shape_class: planar_aromatic
 likely_losses: CH3O; CH3
 fragile_bonds: C-O
 retrieval_terms: dimethoxybenzene; methoxy; phenyl; C15H16O3; 244.1099;
 neutral
 END
 BEGIN
 151
C1=CC(=CC=C1C(CCO)C2=CC=C(C=C2)O)O

 analysis_type: structure_retrieval_signature
 record_id: record_001927
 smiles: C1=CC(=CC=C1C(CCO)C2=CC=C(C=C2)O)O
 exact_mass: 244.1099
 molecular_formula: C15H16O3
 heteroatom_signature: N0 O3 S0 P0 hal0
 hbd: 3
 hba: 3
 xlogp_class: medium
 adduct_compatibility: [M+H]⁺; [M+Na]⁺; [M-H]⁻
 consistency_check: consistent
 scaffold: diphenylmethane
 ring_system: fused_aromatic
 functional_groups: phenol; alcohol
 acid_base_class: weak_acid
 shape_class: planar_aromatic
 likely_losses: H2O; CH3OH

```

fragile_bonds: C-O(phenol); C-O(alkyl)
retrieval_terms: diphenylmethane; phenol; alcohol; C15H16O3; 244.11;
    weak_acid
END
BEGIN
166
CC1=COC=CC2(C1CC3=C(C(=O)OC3=C2)C)C

analysis_type: structure_retrieval_signature
record_id: record_001942
smiles: CC1=COC=CC2(C1CC3=C(C(=O)OC3=C2)C)C
exact_mass: 244.1099
molecular_formula: C15H16O3
heteroatom_signature: N0 O3 S0 P0 hal0
hbd: 0
hba: 3
xlogp_class: medium
adduct_compatibility: [M-H]-; [M+Cl]-; [M+HCOOH-H]-; [M+CH3COOH-H]-
consistency_check: consistent
scaffold: dihydrocoumarin
ring_system: fused_aromatic
functional_groups: lactone; alkene; alkane
acid_base_class: neutral
shape_class: fused_aromatic
likely_losses: CO2; CH3
fragile_bonds: lactone
retrieval_terms: dihydrocoumarin; lactone; fused_ring; neutral; C15H16O3;

```

244.11

END

BEGIN

180

CC1=C2C[C@]3(C(=C)[C@H]4C[C@H]4[C@@]3(C=C2OC1=O)C)O

analysis_type: structure_retrieval_signature

record_id: record_001956

smiles: CC1=C2C[C@]3(C(=C)[C@H]4C[C@H]4[C@@]3(C=C2OC1=O)C)O

exact_mass: 244.1099

molecular_formula: C15H16O3

heteroatom_signature: 03

hbd: 1

hba: 3

xlogp_class: medium

adduct_compatibility: [M+H]⁺; [M+Na]⁺; [M-H]⁻

consistency_check: consistent

scaffold: fused_tricyclic

ring_system: tricyclic_5-6-6

functional_groups: lactone; alkene; alcohol; exocyclic_alkene

acid_base_class: neutral

shape_class: mixed

likely_losses: H2O; CO2

fragile_bonds: lactone; exocyclic_alkene

retrieval_terms: lactone; tricyclic; exocyclic_alkene; alcohol; C15H16O3;
neutral_loss_H2O

END